

Transaction Risk Analysis Using SQL

Project Overview

1. Objective

The objective of this project was to analyse financial transaction data using SQL to identify suspicious transaction patterns and assess potential fraud risks. The analysis focused on detecting behavioural anomalies commonly used in fraud detection and AML monitoring, including unusually high-value transactions, abnormal transaction frequency, threshold avoidance, geographic inconsistencies, and deviations from historical customer behaviour.

2.Dataset

This project utilized a simulated credit card transaction dataset containing customer transaction records. The dataset included customer ID, transaction amounts, transaction timestamps, countries, IP addresses, and behavioural attributes required for transaction risk analysis. The data was used to identify suspicious activity based on common fraud risk indicators.

3. Tools Used

Tool	Purpose
SQL /DB Browser for SQLite	Querying and analysing transaction data
Microsoft Excel	Reviewing and validating dataset outputs
Microsoft Word	Documenting findings and preparing the project report

4. Methodology

Process

Imported the CSV transaction dataset into a SQL database.

Developed SQL queries to analyse customer transaction behaviour.

Applied queries to detect common fraud indicators.

Evaluated query results based on transaction behaviour.

Assigned appropriate risk levels (Low, Medium and High).

Recommended investigation or monitoring actions for each identified risk.

Fraud Risk Indicators Analysed

- High-value transactions
- High-frequency transactions
- Threshold avoidance
- Geographic inconsistency
- Behavioural deviation

5. Key Findings

Indicator	Observation	Risk Level	Action
High-value transactions	25 users exceeded normal patterns; anomaly ratios up to 26x (UK)	High	Flag for manual review
High-frequency	No abnormal bursts detected	Low	Continue monitoring
Threshold avoidance	User CC_9246 repeatedly made transactions near the threshold limit	Medium	Monitor & verify recurring activity
Geographic inconsistency	6 users transacted across multiple countries/IPs	Medium	Trigger authentication & review
Behavioural deviation	2 users spent >2x average	Medium	Flag for verification

The analysis identified several behavioural patterns commonly associated with financial crime. High-value transaction spikes represented the highest risk indicator within the dataset. Geographic inconsistencies and repeated near-threshold transactions suggested potential attempts to bypass automated monitoring systems. Behavioural deviation analysis highlighted users whose spending patterns significantly differed from their historical behaviour, warranting additional review.

6. Business Recommendations

- Implement automated monitoring for high-value transactions.
- Review repeated transactions below predefined thresholds.
- Verify transactions with unusual geographic activity.
- Escalate accounts exhibiting multiple fraud indicators for manual investigation.

7. Skills Demonstrated

Technical: SQL

Data Analysis

Excel

Analytical: Fraud Detection

Transaction Monitoring

Risk Classification

Investigation Planning

8. Conclusion

This project applied SQL to analyse credit card transaction data and identify common fraud risk indicators, including high-value transactions, threshold avoidance, geographic inconsistencies, and behavioural deviations. The analysis helped classify transaction risks and recommend appropriate investigation actions, demonstrating practical application of fraud monitoring and risk assessment.